

## Session 4 — Live demo: the 2×2 calculator

Foundations of Clinical Research · Week 4 · Saturday · Neudata · Instructor script

|                     |                                                                                                                                                                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Slots</b>        | <b>Main demo (8 min):</b> section 4.2, just before the break, slide "Live demo: the 2x2 calculator" (sheet <i>Association</i> ). <b>Short add-on (1–2 min, optional):</b> section 4.3, slide "Same test, different setting: PPV depends on prevalence" (sheet <i>Diagnostic</i> ). |
| <b>File</b>         | Course/Resources/tools/2x2_calculator.xlsx (regenerate with <code>python make_2x2_calculator.py</code> in the same folder)                                                                                                                                                         |
| <b>Software</b>     | Excel or LibreOffice Calc (any recent version). No internet needed.                                                                                                                                                                                                                |
| <b>Participants</b> | Watch only. No laptops needed.                                                                                                                                                                                                                                                     |
| <b>Goal</b>         | Everyone sees risk, RR, RD, NNT and OR computed from ONE table and hears each one in plain words; in 4.3 they see the same test give very different PPVs in two settings.                                                                                                          |

### Before the session (5 minutes, the day before)

1. Open the file and check that it shows the **worked example**: Restrictive 20 / 80, Usual care 30 / 70 (sheet **Association**); 18 / 16 / 2 / 64 and a new-setting prevalence of **2%** (sheet **Diagnostic**). If someone has changed it, rerun `make_2x2_calculator.py`.
2. Set the zoom to about **150%** so the back row can read it. Close every other window.
3. Only the yellow cells can be edited; the sheets are protected without a password (Review → Unprotect Sheet if you need to change a label).
4. Keep a whiteboard marker ready for the fallback.

### Part 1 — Main demo in 4.2 (8 minutes, sheet *Association*)

#### Step 1 — Show the layout (1 minute)

**Do:** open sheet **Association**. Point at the yellow cells.

**Say:** "Same layout as the slide: treatment or exposure in rows, with the treated group on top; outcome in columns, with 'died' on the left. a, b, c, d — always in the same place."

**Appears:** the 2×2 table with 20 | 80 and 30 | 70; totals 100 and 100.

#### Step 2 — Read the results in words (3 minutes)

**Do:** point at each row of the results table, reading the right-hand "In plain words" column aloud.

| Row                       | Value shown | Plain words shown                                           | Teaching point                        |
|---------------------------|-------------|-------------------------------------------------------------|---------------------------------------|
| Risk in exposed / treated | 20.0%       | 20 of 100 patients in the Restrictive group had the outcome | Risk is out of everyone in the group. |

| Row                         | Value shown   | Plain words shown                                    | Teaching point                           |
|-----------------------------|---------------|------------------------------------------------------|------------------------------------------|
| Risk in unexposed / control | 30.0%         | 30 of 100 ... Usual care ...                         |                                          |
| Risk ratio (RR)             | 0.67          | The risk is 33% LOWER in the exposed/treated group   | Relative effect.                         |
| Risk difference (RD)        | -10.0%        | 10.0 fewer patients with the outcome per 100         | Absolute effect — the one patients need. |
| NNT / NNH                   | 10.0          | NNT: treat about 10 patients to PREVENT one outcome  |                                          |
| Odds (two rows)             | 0.25 and 0.43 | Odds compare those WITH the outcome to those WITHOUT | Odds are not risks.                      |
| Odds ratio (OR)             | 0.58          | outcome is common, so OR lies further from 1 than RR | Matches the OR-vs-RR chart.              |

### Step 3 — Make the outcome rare (2 minutes)

**Do:** type **2** in a (Restrictive, died), **98** in b, **3** in c (Usual care, died), **97** in d. Press Enter after each.

**Appears:**

|       | Before     | After                         |
|-------|------------|-------------------------------|
| Risks | 20% vs 30% | 2% vs 3%                      |
| RR    | 0.67       | <b>0.67 (unchanged)</b>       |
| RD    | -10%       | <b>-1%</b>                    |
| NNT   | 10         | <b>100</b>                    |
| OR    | 0.58       | <b>0.66 — now close to RR</b> |

**Say:** "Same relative effect, but now we must treat 100 patients to prevent one death. And because death is now rare, the OR has almost caught up with the RR."

**Do:** type the original numbers back (20, 80, 30, 70).

### Step 4 — Case-control warning (1 minute)

**Do:** point at the red line under the table.

**Say:** "The spreadsheet will happily calculate a 'risk' from case-control data. Do not read it — the researchers chose how many cases and controls to include. In your second calculation, only the OR is valid. After the break you will do two calculations yourselves — Part C of your handout."

## Part 2 — Add-on in 4.3 (1–2 minutes, optional, sheet *Diagnostic*)

Use this only if the laptop is already open and you are on time; the slide makes the same point without it.

1. **Do:** click the **Diagnostic** tab. **Appears:** sensitivity 90.0%, specificity 80.0%, PPV 52.9%, NPV 97.0%, prevalence 20.0% (the ICU picture).
2. **Do:** point at the blue box "Same test, another setting". The yellow prevalence cell shows **2%**. **Appears:** PPV in the new setting **8.4%**, NPV **99.7%**.
3. **Say:** "Same sensitivity, same specificity. Screening a population where 2 in 100 have the disease, fewer than 1 in 10 positive results are real."
4. **Do (optional):** point at the small table underneath: prevalence 1% / 2% / 5% / 20% / 50% gives PPV 4.3% / 8.4% / 19.1% / 52.9% / 81.8%.

## Teaching points to land

1. **One layout, every measure.** Most errors are layout errors, not arithmetic errors.
2. **Relative versus absolute.** An RR of 0.67 can mean an NNT of 10 or of 100, depending on the baseline risk.
3. **OR  $\approx$  RR only when the outcome is rare.**
4. **Case-control  $\rightarrow$  OR only.**
5. **Sensitivity and specificity belong to the test; PPV and NPV depend on how common the disease is.**

## After the calculation exercise — checking the answers (about 8 minutes, after the break)

Type each calculation into the Association sheet in front of the room:

| Exercise (handout Part C)                | Type                     | Shows                               |
|------------------------------------------|--------------------------|-------------------------------------|
| Calculation 1 (mattress trial)           | a 10, b 190, c 20, d 180 | RR 0.50, RD -5.0%, NNT 20, OR 0.47  |
| Calculation 2 (candidaemia case-control) | a 24, b 16, c 16, d 64   | OR <b>6.00</b> (ignore the "risks") |

Discuss each result in words (one sentence each), including why Calculation 2 gives only an OR. Then restore the worked example (20, 80, 30, 70) so the file is ready for the next course.

## Fallback: no projector or no spreadsheet

Draw this on the whiteboard and do it with the room — the numbers are chosen to be easy:

|             | Died                                         | Survived | Total                             |                    |
|-------------|----------------------------------------------|----------|-----------------------------------|--------------------|
| Restrictive | 20                                           | 80       | 100                               | risk 20/100 = 0.20 |
| Usual care  | 30                                           | 70       | 100                               | risk 30/100 = 0.30 |
|             |                                              |          |                                   |                    |
| RR          | = 0.20 / 0.30                                |          | = 0.67                            |                    |
| RD          | = 0.20 - 0.30                                |          | = -0.10 (10 fewer deaths per 100) |                    |
| NNT         | = 1 / 0.10                                   |          | = 10                              |                    |
| OR          | = (20 x 70) / (80 x 30) = 1400 / 2400 = 0.58 |          |                                   |                    |

For the rare-outcome point, write 2/100 vs 3/100: RR = 0.67, NNT = 100, OR = (2 x 97) / (98 x 3) = 194 / 294 = 0.66.

For the PPV point, use the slide "Same test, different setting" (PPV  $90/170 = 53\%$  in the ICU;  $9/107 = 8\%$  in screening).

## Common problems

| Problem                             | Fix                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------|
| Cells show –                        | One of the four counts is empty or zero in a denominator. Type a number in every yellow cell. |
| Cannot type in a cell               | Only yellow cells are editable (protected sheet). Review → Unprotect Sheet (no password).     |
| Formulas show as text (LibreOffice) | Tools → Options → LibreOffice Calc → View: untick "Formulas"; or press Ctrl + `.              |
| Prevalence typed as 2 instead of 2% | Type 2% or 0.02; the cell is formatted as a percentage.                                       |
| Numbers look wrong                  | Check the layout: treated/exposed on top, "died" on the left.                                 |